Gemma 4 31B Transport Autopsy Development/Methods Record

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1 Scope

This isolated side study tests exact local transport validity. It does not open a Phase 4 confirmatory model cell and does not infer workspace absence from readout opacity or transport failure.

2 G1 firewall

Analytic and tiny-model exact-JVP goldens and an in-session OLMo positive control precede every Gemma interpretation. Numeric thresholds are calibrated on the control and committed before the first Gemma target result. Finite differences are labeled secants, never exact JVPs. The finite baseline, perturbation, and exact JVP use identical batch shapes and slots. Diagnostics use separately realized post-cast vectors, and the SNR floor includes the target dtype's local quantization step.

3 Paper-band convention

The primary paper's Methods reindex sampled residual-stream layers to $[0, 100]$, so its L38–L92 workspace range is a 38–92% depth convention. On Gemma's 60 blocks this maps approximately to L23–L55. Thus the proposed late candidates L44/L48/L52 are in the paper-relative band; the 37–62% range in later Gemma material is not the governing convention. A valid late transport gate would license a scoped readout assay, not a workspace claim. This convention is registered as a methods artifact before model execution.

4 Architecture

The pinned Gemma text decoder has 60 layers of width 5,376. A five-local, one-global attention schedule repeats ten times. Global attention uses a different head geometry and K=V; four RMSNorm sites and a gated MLP occur per block. The tied output head is followed by a monotone softcap.

5 State of evidence

The foundation and exact-JVP goldens pass. Both PyTorch autodiff backends hit the analytic derivative exactly; forward-mode, fallback, and an independently materialized reverse Jacobianvector product agree within 8.3×10^{-17} on a nonlinear tiny attention/gated-MLP suffix. Central-secant error decreases quadratically with radius. No Gemma target result is reported at this boundary. The exact OLMo snapshot is fully verified (26 files, 14 shards, 64,476,964,249 bytes, zero failures), and the pre-result batch/delivery repair and paper-band convention are registered. All 56 control cells subsequently checkpointed. A provenance-separating pure finalizer repaired only derived JSON/Parquet scalar storage, loaded no model, and registered all 1,568 rows. All 28 clean suffix checks and exact-JVP primal comparisons are bit-exact. Single-position median tangent cosine rises from 0.693 at L4 to 0.991 at L56 and 0.996 at L60. The prepared target gate requires SNR 20, cosine at least 0.98, forward relative error at most 0.20, and central relative error at most 0.10 at the declared 0.10 residual-norm dose. All 32 late control anchors pass. The positive-control event and byte-identical threshold file are registered before any Gemma number. The exact 12-file, two-shard Gemma snapshot is independently hash-verified and the 40-cell execution manifest is frozen. Stage 1 then completed all 40 cells and 1,120 rows from the same clean commit. All 20 suffix checks and exact-JVP primal comparisons are bit-exact. The primary smallest-high-confidence selection passes 0/12, 0/13, 0/12, 0/13, and 1/14 rows at L22/L30/L37/L44/L52; the frozen rule classifies every layer as local tangent mismatch. At the declared 0.10 dose, 12/16 rows per layer are evaluable and none pass. This is an operational transport-instrument result, not a claim of missing information, nondifferentiability, or workspace absence. The frozen actual-model dual-backend comparison then returned finite results from both backends and reproduced the selected Stage-1 row exactly. However, its stricter comparison across all eight original batch slots had tangent cosine 0.99999958 and relative error 0.002458, exceeding the precommitted 10^{-5} relative-error ceiling. The methods gate therefore fails. Its threshold is not relaxed after observing Gemma, and mechanism localization stops at this boundary. The terminal model-free release verifies all 18 source events and 35 source outputs and publishes a methods-only state of record, claim ledger, transport gate protocol, and Phase-4 import envelope. Its status is `COMPLETE_METHODS_BLOCKER`; it opens no model cell or intervention.